Agri-Logistics IDAS: Trilingual Context-Aware Intelligent Driver Assistance System

EPICS in IEEE Engineering Service-Learning Proposal · Project Summary (Oct 2026 Cycle)

Lead Student Applicant: Lokesh Kumar (IEEE Student Member # 99402233 · ID: 2k22-SE-42), Sindh Agriculture University, Tandojam
IEEE Senior Advisor: Prof. Dr. Bhawani Shankar Chowdhry (Life Senior Member, IEEE · HEC Distinguished National Professor, MUET)
Community Partner: Directorate of Agricultural Extension Services, SAU Tandojam | **Live System:** https://agri-idas.tech

1. Community Need & Partner Collaboration in Rural Sindh

In Lower Sindh, agricultural supply chains suffer over **30% to 40% post-harvest crop loss** (tomatoes, chillies, onions) during transit to terminal wholesale markets. Partnering with the **Directorate of Agricultural Extension Services (SAU Tandojam)** across all phases (co-designing collector corridors, in-field pilot testing, and permanent vehicle stewardship), this project addresses the reality that rural freight drivers speak indigenous languages (**Sindhi** and **Dhatki**) and cannot safely interpret standard text GPS while driving. Furthermore, existing navigation tools rely on naive straight-line routing, ignoring canal diversions and monsoon hazards, triggering fuel waste and delays.

2. Technological Innovation & System Architecture

Agri-Logistics IDAS bridges this digital divide through an edge-responsive, voice-first intelligent co-pilot: **(1) True-Road GIS Engine:** Powered by Open Source Routing Machine (OSRM) graph traversal to track asphalt centerlines across rural Sindh; **(2) Trilingual Speech Synthesis:** Synthesizing auditory hazard and turn guidance in **Dhatki**, Sindhi, and Urdu in < 1.0 s; **(3) Telematics Safety Matrix:** Dynamic speed capping based on real-time surface friction (wet μ=0.38) and perishable temperature exposure; **(4) In-Cab Hardware Unit:** Low-cost edge box (ESP32-S3 microcontroller, GPS transceiver, and offline DAC audio output) suitable for vintage trucks.

3. Empirical Validation Across 8 Regional Corridors (Already Proven)

Empirical Metric	Benchmarked Value	Statistical / Operational Significance
Routing Discrepancy Error	**17.07%** (Up to 25.05%)	Paired t-test: **$t = 4.892$, $p = 0.0017$** ($p < 0.01$). Rectifies naive straight-line models.
Unbudgeted Fuel Deficit	**8.84 Liters / Rs. 2,488**	Fuel saved per fleet cycle across 8 Sindh corridors (@ Rs. 282/L baseline).
Dhatki Semantic Intent Parsing	**0.0034 ms** (Mean)	One-way ANOVA ($F = 0.421$, $p = 0.662$): Zero latency penalty for indigenous dialect.
End-to-End Voice Turnaround	**747.92 ms** (95%: 766 ms)	Conforms strictly to ISO/IEEE automotive safety standard (< 1.0 second response).
Driver Usability Scale Reliability	**Cronbach's α = 0.842**	Pilot survey with 25 commercial truck drivers confirms high construct reliability.

4. Itemized Project Budget & Resource Allocation

Official EPICS Budget Category	Cost (USD)	Line Item Detail & Community Service Justification
Physical Prototype Supplies (Kits)	**$3,800**	25 in-cab ESP32-S3 edge boxes, high-precision NEO-M8N GPS receivers, and amplified DAC speakers.
Physical Prototype Supplies (Mounts)	**$1,400**	12V-to-5V isolated automotive step-down converters, vibration-dampened mounts, and IP65 cab harnesses.
Physical Prototype Supplies (Gateway)	**$1,400**	Dedicated field dispatch edge gateway hardware stationed at SAU Extension center for offline GIS map & telemetry sync.
Travel (Local only — 8 Corridors)	**$1,600**	Local transit fuel & field travel for student engineering team to conduct corridor calibration & telemetry audits.
Event Costs (Driver Training)	**$1,200**	3 community driver training workshops hosted via SAU Extension (Mirpurkhas, Kunri, Tandojam)—hall, AV & orientation.
Logistics (Trilingual Manuals)	**$600**	Printing 150 pictorial laminated driver navigation manuals (Sindhi, Dhatki, Urdu) for 25 fleet trucks.
TOTAL REQUEST FROM EPICS IN IEEE	**$10,000 USD**	**100% EPICS Compliant (Strictly Prototyping, Local Travel, Events, & Logistics; No Excluded Fees)**

5. 12-Month Milestones, Student Learning & Sustainability

- **M1–M3 (Engineering Design):** Hardware fabrication of 25 in-cab units; offline Dhatki/Sindhi phoneme firmware; SAU Extension field co-design.
- **M4–M6 (Pilot Deployment):** In-cab mounting across 25 agricultural freight trucks (Mirpurkhas–Kunri–Tandojam); Q1 check-in report to EPICS in IEEE.
- **M7–M9 (Peak Harvest Trials):** Longitudinal telemetry on fuel savings, transit latency, and cargo spoilage; Q2/Q3 check-in reports.
- **M10–M12 (Handover & Report):** Permanent hardware stewardship handover to SAU Extension Directorate; submission of IEEE Final Technical & Financial Report.
- **Student Learning & Governance:** Students master embedded AI, GIS topology, and community engineering; funds routed via **IEEE Karachi Section**.